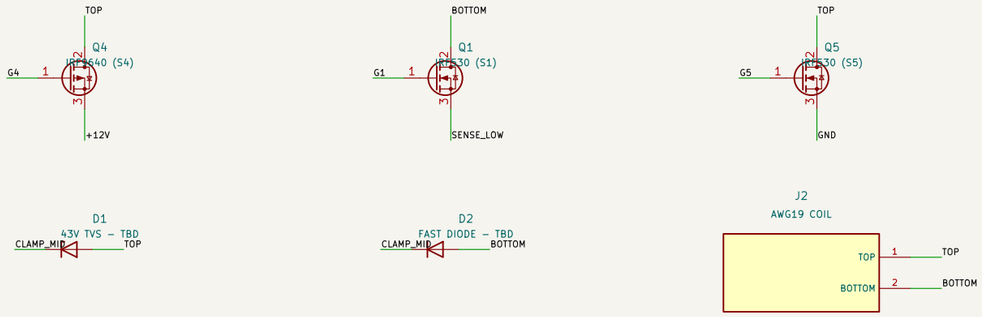
AWG 19 PROTON MAGNETOMETER — POLARIZATION DRIVER + CURRENT MONITOR

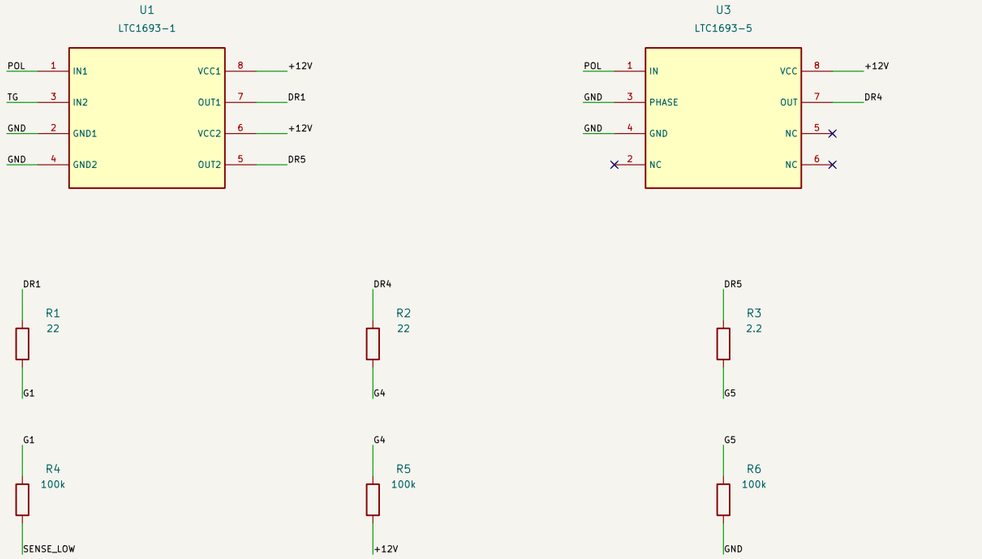
Engineering draft. Named nets connect across blocks. Based on Koehler Revision 4 and the latest LTspice circuit.

1. POWER SWITCHES / EXTERNAL COIL / CLAMP



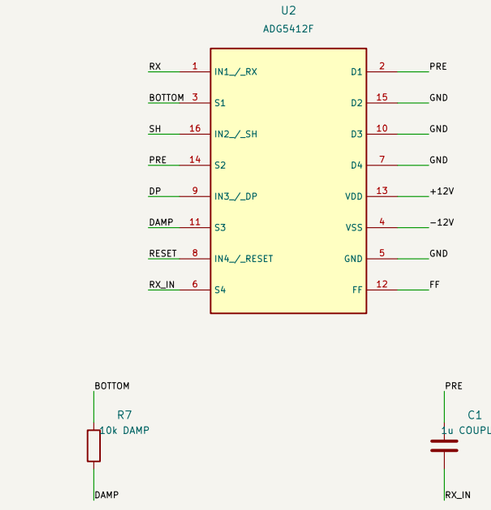
Coil: R=11.430016 ohm, L=25.92669 mH, 1928 turns.
D1 cathode joins D2 cathode. Clamp ratings / exact parts TBD.

2. MOSFET GATE DRIVERS



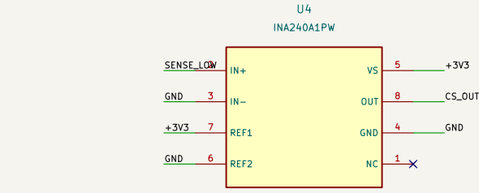
POL=1 turns Q1/Q4 on. TG=1 turns Q5 on.
MCU must enforce dead time. No hardware interlock included.
Use regulated 12 V. Verify MCU VOH against driver VIH.

3. RECEIVER ISOLATION / DAMPING



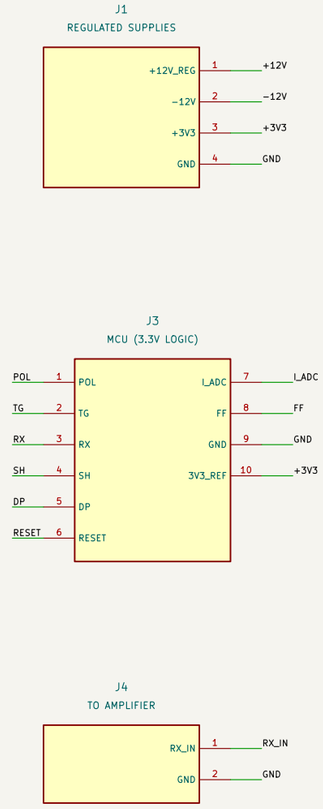
U2 source pins face protected nodes. Four channels, one IC.
RX load belongs to external amplifier. Simulation-only
100k / 10p load and 1n coil parasitic are omitted.

4. POLARIZATION CURRENT SENSOR



Kelvin sense across R9. I = (Vadc − Vzero) / 1 V/A.
Vzero −1.65 V; 1.008 A −> 2.658 V in nominal simulation.
Measures polarization only. Clamp current bypasses R9.

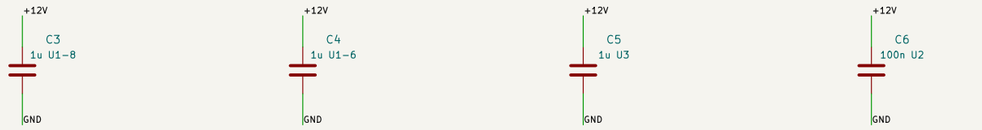
5. EXTERNAL INTERFACES



Simulation sequence (ms):
5–25: polarization
25: turn off, discharge
25.1: TOP grounding
37: damping off
38.98: shunt off
39: receive on
40: input reset off

DRAFT: DO NOT FABRICATE
Clamp parts, footprints, noise, startup defaults and interlock need review. Coil receiver residual −10.87 mV remains.

6. LOCAL SUPPLY BYPASS (PLACE AT IC PINS)



ECE 455 — Jun Yang

Sheet: /
File: Polarization_AWG19.kicad_sch

Title: AWG19 Polarization Driver + Current Sensor

Size: A3	Date:	**Rev: DRAFT A**
KiCad E.D.A. 10.0.6		Id: 1/1